COMMUNITY ANIMAL CARE

PawConnect

Project Viva Questions and Answers

FROM PROJECT BASICS TO DEPLOYMENT

A concise guide to the application, its files, request flow, database, security, and hosting.

Beginner

1. What is PawConnect?

PawConnect is a web application that helps a community coordinate animal-feeding visits, find feeding spots, and share animal welfare reports.

2. What problem does the project address?

Volunteer information can be scattered across messages and informal groups. PawConnect puts locations, schedules, participation, and report status in one shared application.

3. Who uses PawConnect?

Community volunteers browse spots and visits, register, join activities, and submit reports. An administrator can manage spots and report status.

4. What are the main features?

Account registration and sign-in, dashboard counts, searchable feeding spots, scheduled visits, task participation, visit completion, and welfare reports.

5. What technologies does the project use?

React and Vite build the frontend; Java and Spring Boot provide the API; Spring Data JPA and Hibernate access the database; Neon PostgreSQL stores hosted data.

6. What happens when a visitor opens the home page?

Spring's PageController forwards the root route to the built React index. React loads its JavaScript and CSS, then requests dashboard data from the same server's API.

7. What does the dashboard show?

It shows counts of registered volunteers, feeding spots, visits, and reports still pending.

8. Why does the project include sample records?

Sample spots, visits, and a report make the main screens demonstrable on a new database. The seeder does not create an account with a shared password.

9. What is an API?

An API is a defined interface that lets the React frontend send requests to Spring and receive structured data, usually as JSON.

10. What does responsive design mean here?

The layout adjusts to different screen widths so forms, navigation, statistics, and lists remain usable on mobile and desktop.

Intermediate

11. What is React responsible for?

React renders the interactive interface. App.jsx owns UI state and responds to user actions such as searching, signing in, and submitting forms.

12. What is Vite responsible for?

Vite runs the frontend development/build tools. Its production build creates static files under Spring's static/react-build directory.

13. Why is the frontend served by Spring?

Serving both parts from one origin simplifies deployment and lets the browser call the API without a separate cross-origin configuration.

14. What is a REST endpoint?

It is an HTTP route representing an operation or resource. For example, GET /api/spots reads spots and POST /api/spots creates one.

15. What does ApiController do?

ApiController maps API routes to Java methods, checks request data and session permissions, calls repositories, and returns JSON or an HTTP error status.

16. What is a JPA entity?

An entity is a Java class that represents a database table. PawConnect has entities for users, feeding spots, feeding tasks, participants, and animal reports.

17. What is a repository?

A Spring Data repository provides database operations for an entity, such as finding a user by email or saving a report, without handwritten SQL for each basic operation.

18. What does Hibernate do?

Hibernate is the JPA implementation that maps entity objects to SQL statements and database rows.

19. How is a feeding spot created?

React validates required form fields and sends JSON to POST /api/spots. Spring checks the signed-in user and field lengths, assigns the creator, and saves the entity using the repository.

20. How does a volunteer join a visit?

React sends POST /api/tasks/{id}/join. Spring reads the volunteer identity from the session, checks that the visit exists, and records the participant.

21. How does report tracking work?

A signed-in volunteer submits the animal, location, and description. The report starts as PENDING; an administrator can change it to IN_PROGRESS or RESOLVED.

22. How are form errors presented?

The browser marks required fields, while the API independently validates requests. Spring returns a useful status and message, and React shows that message in an alert.

23. What is an HTTP session?

A server-side session stores the signed-in user identity. The browser receives an HTTP-only session cookie and automatically sends it with same-origin requests.

24. Why use BCrypt for passwords?

BCrypt is a one-way password hashing function designed to make password guessing more expensive. PawConnect never needs to store a new account's raw password.

25. How are administrator accounts selected?

The server compares an account email with the ADMIN_EMAIL environment variable. A matching account receives the ADMIN role; other accounts are VOLUNTEER.

26. What is Neon?

Neon is a hosted PostgreSQL service. PawConnect connects to it through the PostgreSQL JDBC driver using server-side environment variables and TLS.

27. Why keep H2 in the project?

H2 is a lightweight local database that lets a student run the project without first creating a cloud database. The Neon profile is used for hosted deployment.

28. How does Spring choose the database?

The default application.properties configures local H2. Setting SPRING_PROFILES_ACTIVE=neon loads application-neon.properties and the Neon JDBC environment variables.

29. What is DemoDataSeeder?

It inserts example spots, visits, and a report when those tables are empty. It does not insert a user with a known password.

30. What does the PageController do?

It maps / and /index to the built React entry page so the application opens at the user interface.

Advanced

31. What is the purpose of the ADMIN role?

It keeps destructive actions and report-status management on the server behind an explicit permission check. Hiding a button in React is not used as the only access control.

32. What happens if an older local account has a plain-text password?

After a successful legacy sign-in, the backend replaces that stored value with a BCrypt hash. Newly registered accounts are hashed immediately.

33. How are secrets kept out of the repository?

The application reads Neon credentials from environment variables. .env files are ignored by Git, and the example file contains placeholders only.

34. What is Docker doing for deployment?

The first image stage installs Node packages and builds React. The second stage packages the Java application with Maven, and the final stage runs the smaller Java runtime image.

35. How does Render use render.yaml?

The blueprint describes the Docker web service, health-check route, Neon profile, and names of secret values that must be supplied through Render.

36. What does the health check verify?

Render requests GET /api/stats. A successful response shows that Spring is serving requests; database connectivity is exercised by the repository-backed counts.

37. Why is sslmode=require in the Neon JDBC URL?

It requires an encrypted connection between the Spring server and the hosted PostgreSQL database.

38. What if Neon is unavailable?

Database-backed requests can fail until connectivity returns. The API returns a readable error and logs diagnostic details on the server; verify the URL, credentials, network, and Neon service status.

39. What are current limitations?

The API loads lists without pagination, has no rate limiting, and does not yet include automated integration tests. A larger public service should add those controls and monitoring before expansion.

40. How would you improve the project next?

Add automated API and browser tests, pagination, rate limiting, audit history for administrator changes, and monitoring for database and application health.

41. How do you run the project locally?

Install JDK 17, Maven, and Node.js; run npm ci and npm run build in frontend; then run mvn spring-boot:run from the project root and open http://localhost:8080.

42. What should be checked before a viva demonstration?

Confirm the app starts, registration and sign-in work, the session persists after refresh, a volunteer can create and join a visit, an administrator can update a report, and hosted data remains after redeployment.